

9.4% Expected return (1y, ex-ante, KRW)	6.1% Volatility (benchmark* 8.7%)	1.07 Sharpe vs 2.8% KRW cash (benchmark* 0.41)	0.28 / 0.11 Beta to S&P 500 / KOSPI 200 (in KRW)	0.7% 95% 1-yr VaR (benchmark* 7.9%)
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Holdings, weights and rationale

Holding	Sleeve	Weight	Why it is held	Valuation / return source
United States · 22 holdings		52.0%		
IVV iShares Core S&P 500 ETF	Growth	5%	Core US beta; mega-caps at index weight	Earnings yld + growth
UNH UnitedHealth Group	Growth	2%	Defensive compounder at trough margins	10-yr FCFF DCF +52%
CI Cigna Group	Growth	2.25%	PBM + insurer at 8x forward P/E	2-stage FCFE +55%
CMCSA Comcast	Growth	1.25%	13% FCF yield, priced for decline	Declining-perpetuity FCFE +24%
VZ Verizon	Growth	2%	Low-beta cash cow, ~6% dividend	Zero-growth FCFE +19%
PYPL PayPal	Growth	1%	12% FCF yield + heavy buybacks	Low-growth FCFE +40%
ADBE Adobe	Growth	1%	Priced as if AI destroys it	10-yr FCFF DCF +29%
USMV iShares MSCI USA Min Vol Factor ETF	Growth	1%	Defensive US equity (low-vol factor)	Low-vol premium
T AT&T	Growth	2.5%	10% FCF yield, low beta	Zero-growth FCFE +40%
XLU Utilities Select Sector SPDR ETF	Growth	1%	Defensive equity; AI power demand	Defensive yield
O Realty Income	Real asset	1.5%	Net-lease income, 5.9% yield	2-stage AFFO discount +17%
VICI VICI Properties	Real asset	1.5%	Experiential net lease at 9.4x AFFO	2-stage AFFO discount +41%
VNQ Vanguard Real Estate ETF	Real asset	1%	Broad US REITs beyond net lease	Earnings yld + growth
IEF iShares 7-10Y Treasury ETF	Hedge	2.5%	Recession / deflation hedge	YTM ≈ 5.0%
VTIP Vanguard Short-Term TIPS ETF	Hedge	3.25%	Inflation hedge, short duration	Real yld + CPI
SGOV iShares 0-3M T-Bill ETF	Hedge	3.75%	USD cash; dry powder	≈4.2%
IAU iShares Gold Trust	Hedge	1.5%	Geopolitics / Fed-credibility hedge	Diversifier
DBMF iMGP DBi Managed Futures ETF	Hedge	4%	Trend-following; stocks & bonds down	Trend premium
TAIL Cambria Tail Risk ETF	Hedge	4.5%	Crash insurance (puts + T-notes)	Convexity
BTAL AGF US Market Neutral Anti-Beta ETF	Hedge	4.5%	Anti-beta: gains when high-beta falls	Negative beta
PDBC Invesco Optimum Yield Diversified Commodity ETF	Hedge	1.5%	Oil / commodity-shock hedge	Roll + collateral
KMLM KFA Mount Lucas Managed Futures ETF	Hedge	3.5%	Second trend model (diversifies DBMF)	Trend premium
Korea · 19 holdings		36.0%		
069500 KODEX 200	Growth	2%	Core Korea beta	Earnings yld + growth
005935 Samsung Electronics Pref.	Growth	1%	AI memory at a 22.5% pref. discount	Cycle DCF +15%
105560 KB Financial Group	Growth	1.75%	Value-up bank, ROE > cost of equity	Justified P/B = +20%
086790 Hana Financial Group	Growth	2%	Value-up bank at 0.76x book	Justified P/B = +29%
000270 Kia	Growth	2%	6.6x P/E, 5.7% dividend	Justified P/B = +43%
005380 Hyundai Motor	Growth	1.25%	Below book on depressed ROE	Justified P/B = +15%
033780 KT&G	Growth	3%	Defensive staples, 14% ROE	Justified P/B = +28%
015760 Korea Electric Power	Growth	3%	Utility at 0.41x book	Justified P/B = +118%
055550 Shinhan Financial Group	Growth	1.75%	Value-up bank below book	Justified P/B = +24%
024110 Industrial Bank of Korea	Growth	3%	Policy bank at 0.43x book	Justified P/B = +101%
005830 DB Insurance	Growth	3%	Non-life insurer at 0.8x book	Justified P/B = +101%
086280 Hyundai Glovis	Growth	2.5%	Logistics, 15% ROE	Justified P/B = +43%
329200 TIGER REITs & Real Estate Infra	Real asset	2.25%	K-REITs & infra, ~9.7% distribution	Distribution ≈9.7%
152380 KODEX 10Y KTB Futures	Hedge	1.5%	KRW duration hedge	YTM ≈ 4.4%
261240 KODEX USD Futures	Hedge	1%	Extra USD vs KRW hedge	FX (carry -1.2%)
459580 KODEX CD Rate Active	Hedge	1%	KRW cash; liquidity	CD rate
132030 KODEX Gold Futures (H)	Hedge	1.5%	Gold, FX-hedged (pure gold exposure)	Diversifier
114260 KODEX 3Y KTB	Hedge	1.5%	Short KRW duration	YTM ≈ 3.3%
114800 KODEX Inverse (KOSPI 200)	Hedge	1%	Direct Korea crash hedge	Negative beta
Japan · 5 holdings		12.0%		
1306 NEXT FUNDS TOPIX ETF	Growth	2%	Core Japan beta + yen	Earnings yld + growth
TM Toyota Motor (NYSE ADR)	Growth	1.75%	Auto leader at 10.8x fwd P/E; NYSE ADR	Justified P/E +24%
9432 NTT	Growth	3%	Low-beta telco, 3% yield	Justified P/E +39%
1343 NEXT FUNDS REIT Index ETF	Real asset	2%	J-REITs, rent reflation	Distribution ≈4.5% (est.)
2561 iShares Core Japan Govt Bond ETF	Hedge	3.25%	Yen safe haven + JGB carry	Yen + ≈2.5% yield
Total		100.0%		

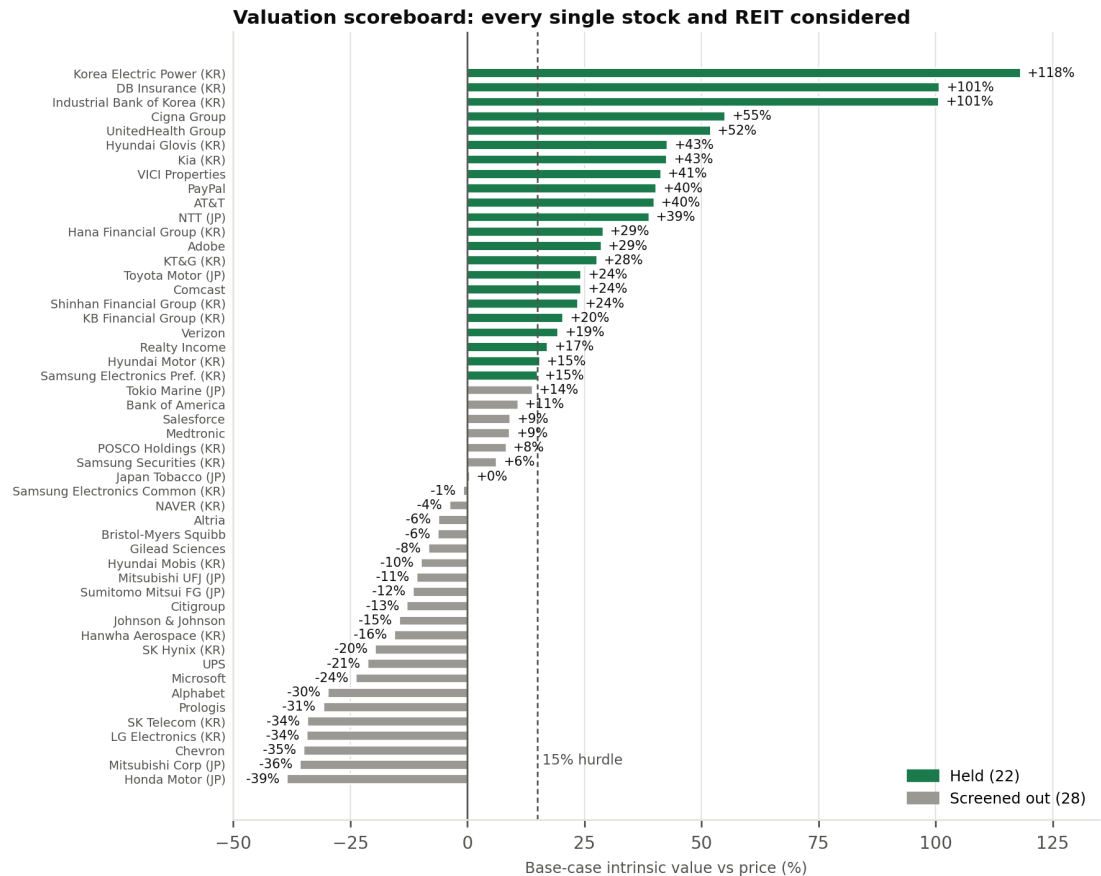
*Benchmark: 30% IVV / 30% KODEX 200 / 20% IEF / 20% KODEX KTB10, in KRW. Expected returns are model outputs, not forecasts; methodology on pages 2, 7 and 9.

1 · Thesis, regime and selection rule

Thesis. (1) With the US 10-year at 5.23% and the JGB 10-year at a 30-year high of 3.07%, discount rates are high, so single-stock risk goes only where a conservative valuation still shows $\geq 15\%$ upside: in practice cheap cash-flow businesses in the US, Korea's 'Value-up' financials and autos, and Japanese names below fair value. (2) Hedges are split by type of crisis (duration, inflation, gold, trend-following, crash insurance) and by currency: for a KRW investor, both the dollar and the yen have historically risen when Korean assets fall. (3) Weights come from a constrained max-Sharpe optimiser with a risk-parity penalty; page 8 tests whether that adds anything over simpler methods.

Indicator (late Sep 2026)	Level	Implication
US 10-yr / 2-yr / Fed funds	5.23% / 4.91% / 4.00%	High discount rate; hikes being priced. Long-duration equity must clear ~9.5–10% cost of equity.
Korea 10-yr KTB	4.41% (+1.56pp y/y)	Korean cost of equity 7–10.5% depending on beta; banks and autos still clear it at today's multiples.
S&P PMI / WTI / Gold	58.4 / \$94 / \$4,266	Overheating + Middle-East oil risk: the nearer tail is inflation, not deflation → hedges must work when bonds fail.
JGB 10-yr / BOJ policy rate	3.07% / 1.25%	Highest JGB yield since 1996; BOJ hiking roughly quarterly. Yen likely to strengthen: Japanese assets carry a yen hedge for a KRW investor.
Samsung / SK Hynix	₩286,500 / ₩1.86m	Korea is riding an AI-memory boom; the job is to own Korea without paying peak-cycle prices.

Rule. Every single stock and REIT gets a cash-flow model matched to its economics (FCFF/FCFE DCF, normalised-cycle DCF, justified P/B for banks and asset-heavy firms, justified P/E for Japan, AFFO discount for REITs). CAPM cost of equity: US 5.23% + $\beta \times 4.5\%$; Korea 4.41% + $\beta \times 5.5\%$; Japan 3.07% + $\beta \times 5.0\%$. **A name is held only if the base case shows $\geq 15\%$ upside** with deliberately conservative inputs. Expected return = cost of equity + closing only **half** of the gap over 3 years, so the optimiser cannot over-trust any one model. ETFs fill the roles single stocks cannot: core beta, duration, inflation, commodities, currency and crash convexity.



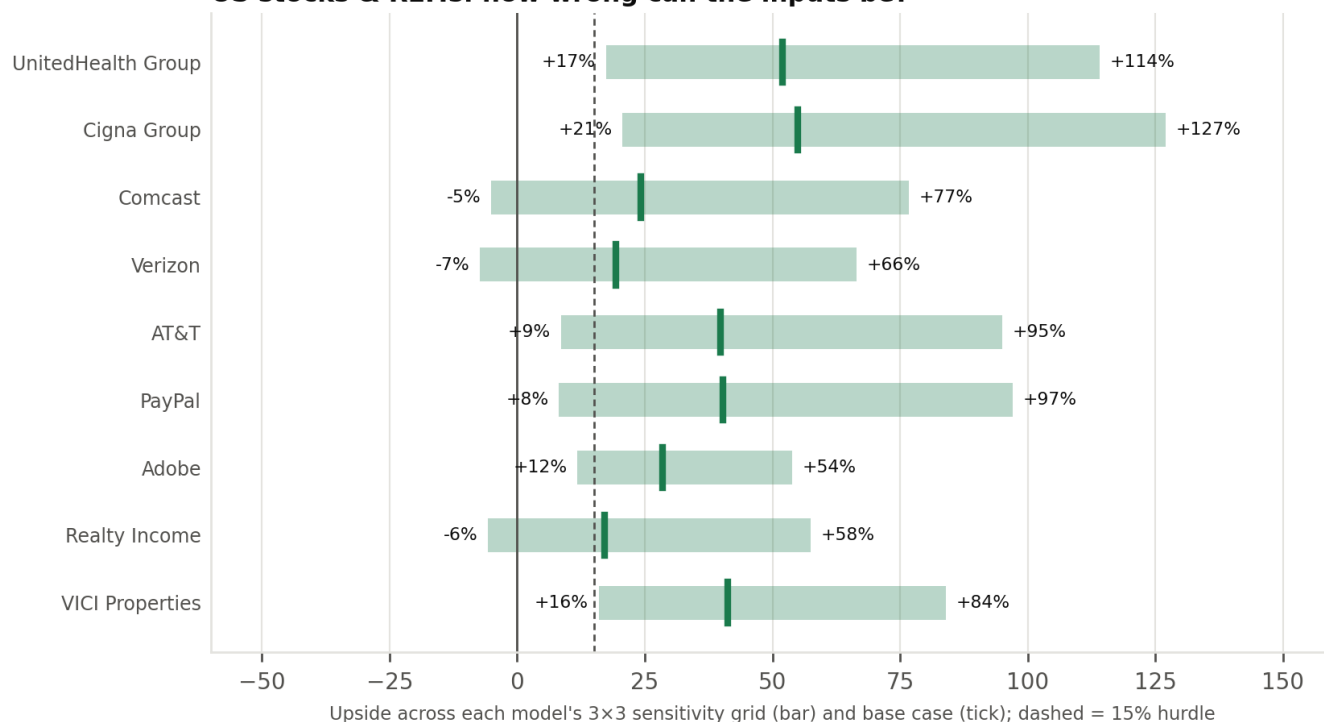
Rejections are part of the thesis. Microsoft and Alphabet price in a WACC of 8.2%–8.3% versus the ~9.5–10% that a 5.23% risk-free rate implies, so they are held only at index weight inside IVV. SK Hynix (-20%) and Samsung common (-1%) are priced near peak-cycle value; Samsung's preferred share, with the same cash-flow rights at a 22.5% discount, is the way in. In Japan, the megabanks (MUFG -11%, SMFG -12%) have already re-rated on BOJ hikes, and Tokio Marine (+14%), Salesforce (+9%) and Bank of America (+11%) are near-misses.

2 · US single stocks and REITs (9 of 22 US holdings)

Holding	Wt	Model	Key inputs	Price	Value	Upside
UnitedHealth Group	2%	10-yr FCFF DCF	Rev \$460bn; FCF margin 4.5%→5.5% (2019-23 avg ≈7%); WACC 8.2%, g 3%	\$376.59	\$572.01	+52%
Cigna Group	2.25%	2-stage FCFE	FCF \$8.0bn (11% yield); +4%/yr 5 yrs then 0% forever; Ke 8.4%	\$274.47	\$425.44	+55%
Comcast	1.25%	Declining-perpetuity FCFE	FCFE \$12.7bn (13.4% yield); -2%/yr forever; Ke 8.6%	\$21.91	\$27.22	+24%
Verizon	2%	Zero-growth FCFE	FCFE ≈\$18bn (OCF \$37.1bn – capex); 0% growth; Ke 7.3%	\$47.08	\$56.21	+19%
AT&T	2.5%	Zero-growth FCFE	2026 FCF guidance \$18bn+ (10% yield); 0% growth; Ke 7.3%	\$25.38	\$35.49	+40%
PayPal	1%	Low-growth FCFE	FCF \$5.6bn FY25 (12% yield); g 1.5%; Ke 10.5%	\$55.04	\$77.20	+40%
Adobe	1%	10-yr FCFF DCF (AI-disruption case)	Rev \$26.6bn; growth 8%→0%; FCF margin 38%→30%; g 1%; WACC 10.2%	\$240.69	\$309.40	+29%
Realty Income	1.5%	2-stage AFFO discount (organic growth only)	AFFO \$4.45 (P/AFFO 12.5x); g 2%→1.5%; k 8.6%	\$55.41	\$64.89	+17%
VICI Properties	1.5%	2-stage AFFO discount (organic growth only)	AFFO \$2.51 (P/AFFO 9.4x); g 2%→1.5%; k 9.3%	\$23.71	\$33.51	+41%

UnitedHealth	FY25 operating margin collapsed to 4.2% on medical-cost inflation. The model assumes only partial repair (FCF margin 5.5% by year 10 vs ~7% in 2019–23) and still finds large upside; the market is using a 10.4% discount rate, i.e. pricing a policy-risk premium.
Cigna	8.2x forward earnings and an ~11% FCF yield. To price PBM (pharmacy-benefit) reform explicitly, the model grows cash flow 4% for five years and then zero forever .
Comcast	Broadband subscriber losses are real, so the model assumes free cash flow shrinks 2% a year forever. At a 13.4% FCF yield, even that is worth 24% more than the price.
Verizon	Zero-growth FCFE of ~\$18bn at a 7.3% cost of equity is worth \$56.21. Beta ≈0.4 and a ~6% dividend make it a bond-like equity; the risk is ~\$190bn of debt.
AT&T	2026 free-cash-flow guidance of \$18bn+ is a ~10% yield. Zero growth at a 7.3% cost of equity is worth \$35.49. With Verizon, telecom is capped at 5% combined because both carry heavy debt.
PayPal	9x forward P/E; FY25 buybacks cut the share count ~7%. Held small because beta is 1.2 and branded-checkout share loss is the live risk.
Adobe	The market prices generative AI as destroying Adobe's moat. The model agrees partly: growth fades to zero and FCF margin falls from 38% to 30%. Still +28%. Held small because this is the highest-uncertainty call.
REITs	Valued on AFFO with organic growth only (2% → 1.5%). VICI survives a permanent 15% AFFO cut (tenant stress) at \$28.48, still +20%. Prologis is rejected at 21x FFO.

US stocks & REITs: how wrong can the inputs be?

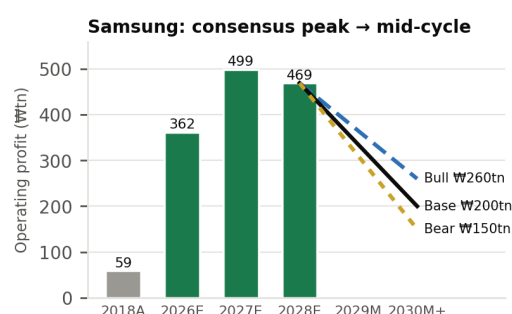


Concentrations: healthcare (UNH + CI = 4.25%) and telecom (VZ + T = 4.5%, capped at 5%). US tech exposure comes mainly through IVV and Adobe. Gilead (-8%) and Medtronic (+9%) were screened and rejected.

3 · Korean single stocks (11 of 19 Korean holdings)

Holding	Wt	Model	Key inputs	Price	Value	Upside
Samsung Electronics Pref.	1%	Normalised-cycle DCF, -10% no-vote haircut	OP ₩499tn/469tn (27-28E) → mid-cycle ₩200tn; WACC 10.5%	₩222,000	₩255,294	+15%
KB Financial Group	1.75%	Justified P/B = (ROE-g)/(k-g)	ROE 10.7% (P/B+P/E); k 9.4%; g 3.0%	₩173,200	₩208,513	+20%
Hana Financial Group	2%	Justified P/B = (ROE-g)/(k-g)	ROE 9.2% (P/B+P/E); k 9.4%; g 3.0%	₩133,100	₩171,676	+29%
Shinhan Financial Group	1.75%	Justified P/B = (ROE-g)/(k-g)	ROE 9.8% (P/B+P/E); k 9.4%; g 3.0%	₩109,300	₩135,119	+24%
Industrial Bank of Korea	3%	Justified P/B = (ROE-g)/(k-g)	ROE 7.4% (P/B+P/E); k 8.3%; g 2.0%	₩20,400	₩40,932	+101%
DB Insurance	3%	Justified P/B = (ROE-g)/(k-g)	ROE 10.9% (P/B+P/E); k 7.7%; g 2.5%	₩184,600	₩370,687	+101%
Kia	2%	Justified P/B = (ROE-g)/(k-g)	Normalised ROE 10% (trailing 10.8%); k 9.9%; g 2.5%	₩118,500	₩168,929	+43%
Hyundai Motor	1.25%	Justified P/B = (ROE-g)/(k-g)	Normalised ROE 9% (trailing 6.5%); k 9.9%; g 2.5%	₩357,000	₩412,050	+15%
Hyundai Glovis	2.5%	Justified P/B = (ROE-g)/(k-g)	ROE 15.3% (P/B+P/E); k 9.4%; g 3.0%	₩196,900	₩281,023	+43%
KT&G	3%	Justified P/B = (ROE-g)/(k-g)	ROE 14.1%; k 7.2%; g 2.5%	₩174,300	₩222,502	+28%
Korea Electric Power	3%	Justified P/B = (ROE-g)/(k-g)	Normalised ROE 7% (trailing 15.9%); k 7.7%; g 1.0%	₩30,250	₩65,974	+118%

Samsung Electronics Pref. Consensus operating profit is ₩499tn for 2027, but memory is cyclical. The model uses consensus for 2027–28 only, then fades to a mid-cycle ₩200tn (still ≈3.4x Samsung's 2018 record) and puts terminal value on that. On this basis the common is fairly valued (-1%). The preferred share has the same cash-flow rights but trades 22.5% cheaper; after a 10% haircut for no votes it clears the hurdle. Bear case (₩150tn mid-cycle): value ₩217,524 for the pref, about the current price.



Banks	KB, Hana, Shinhan and IBK. Justified P/B = (ROE - g)/(k - g); they earn 7–11% on equity against 8–9.4% costs of equity yet mostly trade below book. Korea's 'Value-up' programme raises payouts as P/B approaches 1x. IBK's +101% reflects state-owned-bank cheapness, so treat it with caution. Banks are capped at 9% combined (now 8.5%) and share a 'Korean financials' risk factor.
DB Insurance	10.9% ROE at 0.8x book. IFRS 17 makes insurer book values less comparable, so the +101% is an upper bound and the weight is capped.
Kia & Hyundai	Both trade below book after US-tariff earnings damage. Kia is valued on a normalised 10% ROE (below its trailing 10.8%) and pays a 5.7% dividend. Hyundai's trailing ROE is only 6.5%; on a normalised 9% it barely clears (+15%), so it gets less weight. A shared 'autos' factor in the risk model stops the optimiser treating them as independent.
Hyundai Glovis	Group logistics and car-carrier shipping; 15% ROE at 1.35x book and 8.9x earnings (+43%). Shares an autos factor with Kia and Hyundai.
KT&G	14% ROE, β≈0.5, 3.8% dividend: a defensive compounder. Justified P/B 2.5x vs 1.95x today.
KEPCO	The largest model upside (+118%) and the widest error band. Trailing ROE of 15.9% is flattered by cheap fuel, so the model uses 7%. Tariffs are set politically and debt/equity is 417%, so weight is capped at 3%. It loses in an oil spike (modelled at -25%), which PDBC and DBMF offset.

Rejected in Korea: Samsung Securities (+6% on a normalised 11% ROE; this year's 19% is a trading boom), POSCO Holdings (+8%), plus the names on page 2.

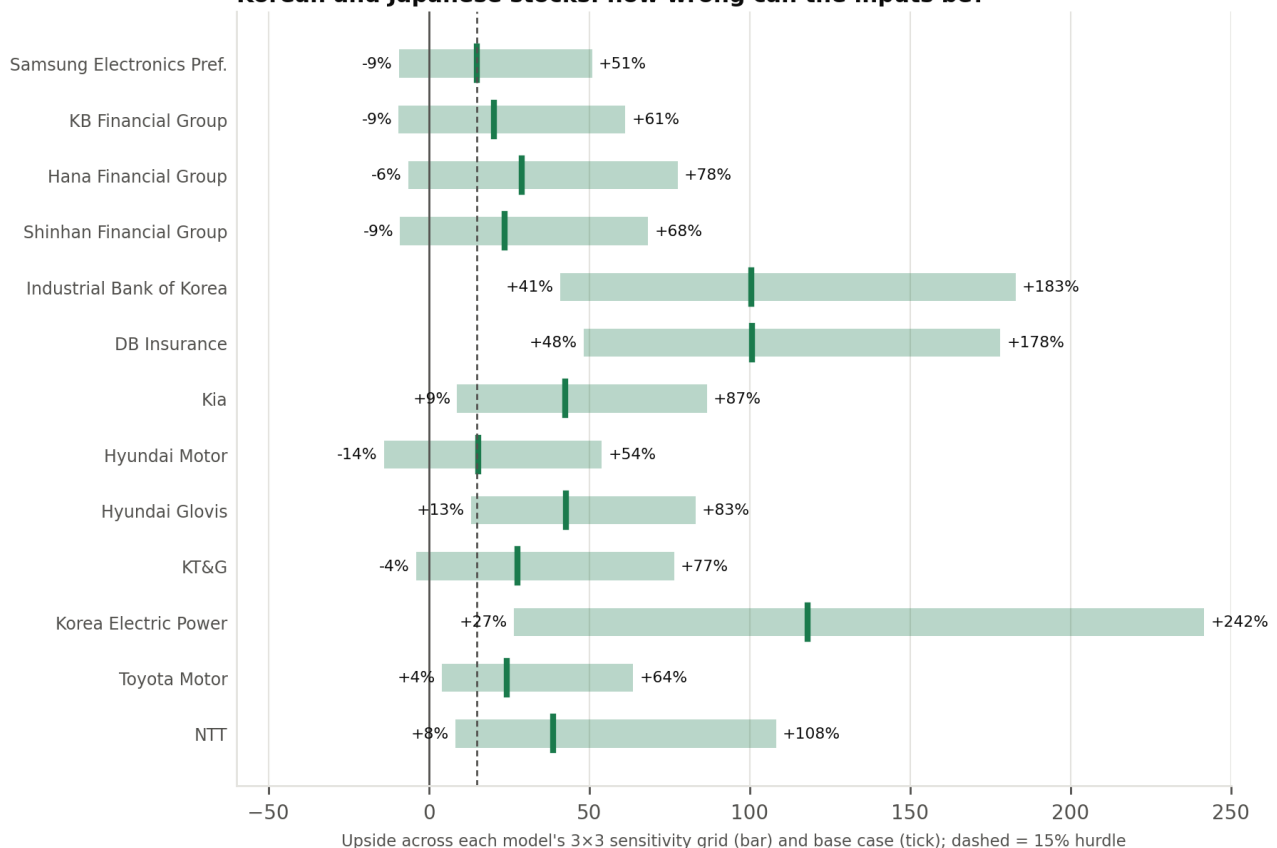
4 · Japan

Why Japan. Three reasons. First, **the yen**: it is a funding and safe-haven currency, so in a global sell-off it tends to rise, and it rises even more against the won. In 2008 JPY/KRW went up roughly 60%. For a Korean investor, yen assets are a second currency hedge next to the dollar. Second, **corporate reform**: the Tokyo Stock Exchange is pressing sub-book companies to raise returns, the same idea as Korea's Value-up but further along. Third, **BOJ normalisation**: with policy at 1.25% and the 10-year JGB at 3.07%, JGBs finally pay a yield.

Holding	Wt	Model	Key inputs	Price	Value	Upside
Toyota Motor	1.75%	Justified P/E (ROE-based payout)	Fwd EPS ¥280, ROE 10%; g 2%; k 8.1%	¥3,025	¥3,758	+24%
NTT	3%	Justified P/E (ROE-based payout)	Fwd EPS ¥12.9, ROE 11%; g 1%; k 6.1%	¥168.8	¥234.3	+39%

Toyota	10.8x forward earnings, 3.3% dividend and active buybacks. Valued with a justified P/E: payout = 1 - g/ROE on a 10% ROE and 2% growth, at an 8.1% cost of equity, gives ¥3,758. Risk: US tariffs and a stronger yen both cut earnings. Held through the NYSE ADR (TM, 1 ADR = 10 shares, ~\$192) : Tokyo trades in 100-share lots (~¥2.6m), which would force a portfolio of ~¥150m to hit a 1.75% weight. The ADR is priced in dollars but tracks the Tokyo share times USD/JPY, so the economic exposure, including the yen, is the same.
NTT	Low-beta telecom at 13x earnings and a 3.1% yield. At a 6.1% cost of equity (β 0.6) its value is ¥234.3. Its main job here is stable yen exposure.
TOPIX ETF (1306)	Core Japan beta across ~2,000 stocks, rather than betting on individual reform stories.
J-REIT ETF (1343)	Japanese REITs: rents are finally rising with inflation. Adds a third real-estate market next to US and Korean REITs.
JGB ETF (2561)	Japanese government bonds. The yield is modest, but in KRW terms this is mostly a yen position: it gains in the crises where the won falls and the yen rises.
Rejected	MUFG (-11%) and SMFG (-12%): already re-rated. Tokio Marine (+14%): a near-miss. Mitsubishi Corp (-36%), Japan Tobacco (+0%) and Honda (-39%): priced at or above fair value.

Korean and Japanese stocks: how wrong can the inputs be?



5 · ETFs: core beta, real assets and a 39.7% hedge sleeve

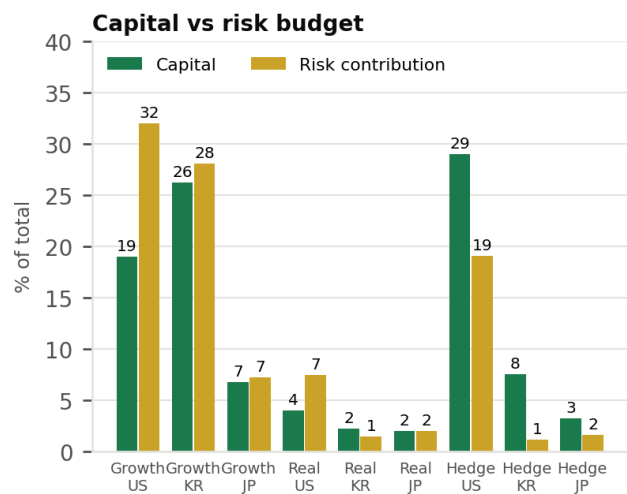
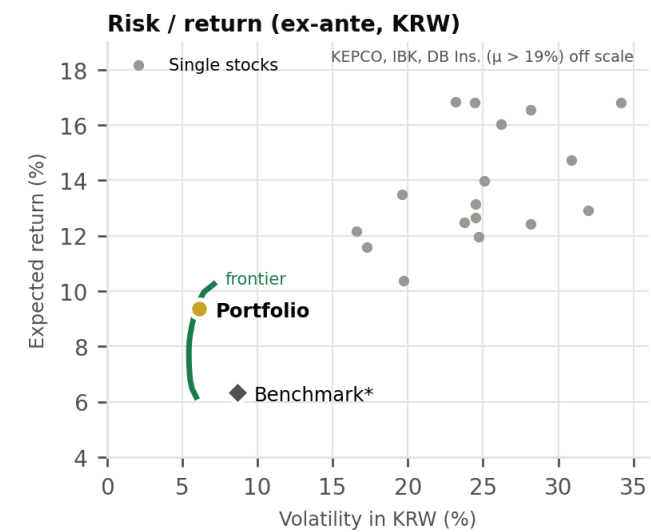
The hedge design is **one instrument per type of crisis**, because 2022 proved that bonds alone fail when inflation is the shock. For a KRW-based investor there are two further hedges: the dollar and the yen. In every Korean crisis on record (1997, 2008, 2020) the won fell as stocks fell, so the 64% of the book held in USD and JPY assets rises in KRW exactly when Korean assets drop.

ETF	Wt	Exposure	Protects against / role	Carry	When it fails
IVV	5%	Core US equity	—	5.0–6.5%	US bear market
USMV	1%	Lower-beta US equity	Sharp sell-offs (falls ~70% as much)	≈6%	Momentum / tech rallies
XLU	1%	US utilities	Defensive; rate-sensitive income	≈6.5%	Sharp rate rises
KODEX 200 069500 · KR	2%	Core Korea equity	—	≈8.5%	Korea-specific shock
TOPIX ETF 1306 · JP	2%	Core Japan equity	—	≈7.5%	Strong yen hurts exporters
VNQ	1%	Broad US REITs	Inflation via rents	≈7%	Rate spikes
J-REIT ETF 1343 · JP	2%	Japanese REITs	Rent reflation	≈4.5% distr. (est.)	JGB yield spikes
TIGER REITs 329200 · KR	2.25%	Korean REITs & infrastructure	Inflation via rents / tolls	≈9.7% distr.	KRW rate spikes
IEF	2.5%	7–10y US Treasuries	Recession, credit crash (+18% in 2008)	≈5.0% YTM	Inflation shock (2022: ~15%)
KODEX KTB10 152380 · KR	1.5%	10y Korean govt bond futures	Korean recession / BOK cuts	≈4.4%	KRW rate rises
KODEX KTB3 114260 · KR	1.5%	3y Korean govt bonds	Low-duration KRW ballast	≈3.3%	BOK hikes (mildly)
JGB ETF 2561 · JP	3.25%	Japanese govt bonds	Yen safe-haven + duration	≈2.5% + yen	Yen weakness
VTIP	3.25%	0–5y US TIPS	Unexpected inflation, little duration	Real yld + CPI	Disinflation
PDBC	1.5%	Diversified commodities	Oil / Middle-East spike (+~20% in 2022)	Collateral + roll	Global recession
IAU	1.5%	Gold	Geopolitics, Fed credibility, USD debasement	None	Rising real yields
KODEX Gold(H) 132030 · KR	1.5%	Gold futures, FX-hedged to KRW	Gold without adding more USD	~1.4% hedge cost	Rising real yields
DBMF	4%	Managed-futures (trend)	Stocks & bonds falling together (CTA index ≈+20% in 2022)	T-bill + trend	Choppy, trendless markets
KMLM	3.5%	Managed futures (different trend model)	Diversifies DBMF's model risk	T-bill + trend	Choppy, trendless markets
TAIL	4.5%	Put options + T-notes	Crash convexity (sudden ~20% equity)	Negative (option bleed)	Slow grind lower; rallies
BTAL	4.5%	Long low-beta / short high-beta	Speculative-growth unwind; $\beta \approx -0.5$	≈Cash	Junk / high-beta rallies
KODEX Inverse 114800 · KR	1%	Short KOSPI 200 (daily inverse)	Korea-specific crash	≈ -3% (short equity)	Any Korean rally; daily reset
KODEX USD 261240 · KR	1%	Long USD vs KRW futures	KRW crisis	≈ -1.2% (rate gap)	KRW appreciation
SGOV	3.75%	0–3m US T-bills	Rate hikes; funds rebalancing	≈4.2%	Rapid Fed cuts
KODEX CD 459580 · KR	1%	KRW CD-rate fund	KRW liquidity	≈2.8% (est.)	BOK cuts

Why hedges get 39.7% of capital but only 22% of the risk. Most hedges are low-volatility and load negatively (blue in the factor map, page 9) on equity or positively on factors that rise in a crisis: duration, gold, USD and JPY vs KRW. The diversification ratio is **2.93** (weighted-average asset volatility ÷ portfolio volatility). The price is paid in bull markets: TAIL bleeds option premium, BTAL loses in speculative rallies, and a strong won hurts every USD and JPY asset (page 7).

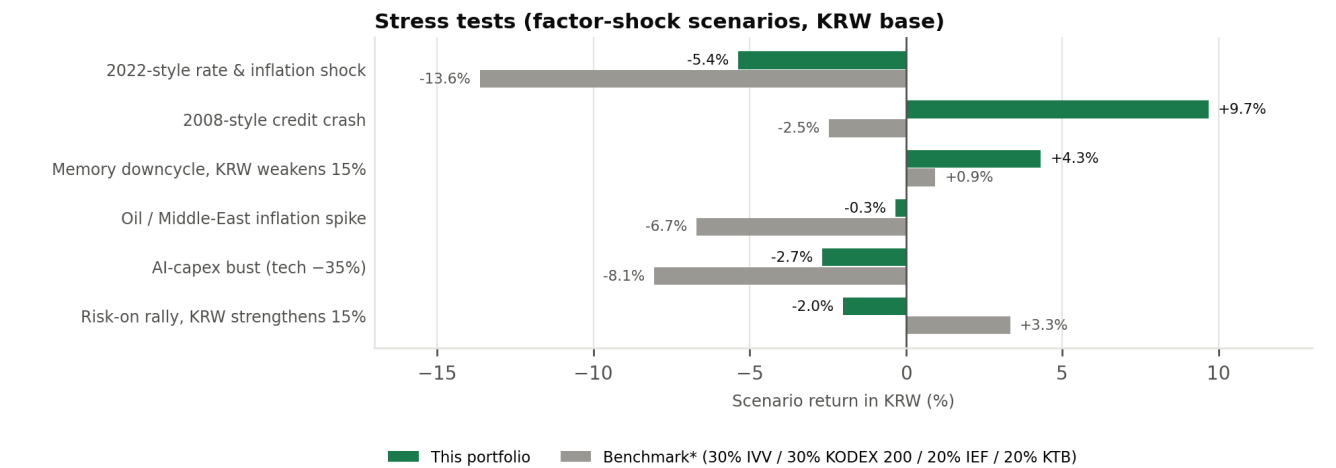
6 · Portfolio construction and risk

Optimiser. Maximise Sharpe ratio over KRW cash with a risk-parity penalty: $\min -SR(w) + \lambda \cdot N \cdot \Sigma(RC_i - \text{mean } RC)^2$, $\lambda = 0.5$. **Constraints:** long-only; single stocks 0.8–3%; REITs 1–3%; IVV 5–12%, KODEX 200 and TOPIX 2–6%; other funds 1–7%; growth sleeve 45–52%, real assets 6–12%, hedges ≥38%; Korea 25–36%, Japan 6–12%; Korean banks ≤9%, telecoms ≤5%. Covariance comes from a 10-factor model (page 9), which keeps a 46×46 matrix stable. SLSQP, rounded to 0.25%.



Ex-ante (KRW)	Portfolio	Benchmark*
Expected return	9.4%	6.3%
Volatility	6.1%	8.7%
Sharpe (vs 2.8% KRW cash)	1.07	0.41
95% VaR / CVaR (1 yr)	0.7% / 3.3%	7.9% / 11.6%
Holdings / largest weight	46 / 5.0%	4 / 30%

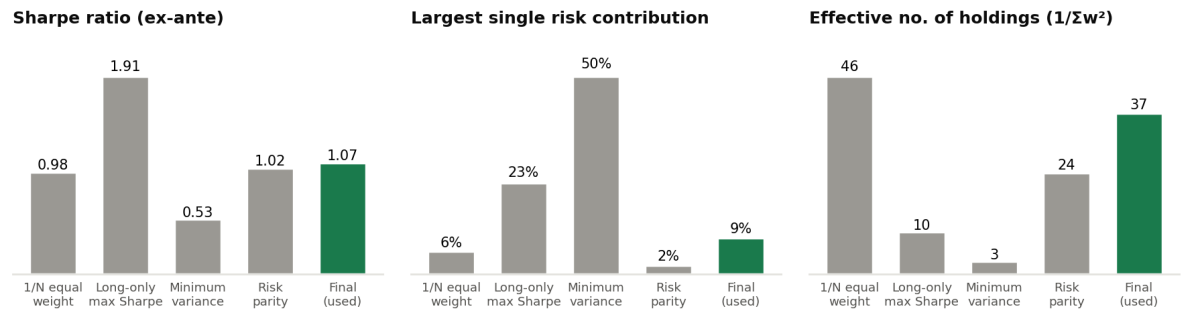
Reading the results. The portfolio holds up in every crisis scenario, and in a 2008-style crash it *gains* in KRW because the won collapses while dollar and yen assets rally. That result depends on those currency relationships holding, which they have in every Korean crisis since 1997 but is not guaranteed. **The cost shows up in the last row:** a risk-on rally with a 15% stronger won returns -2.0% versus +3.3% for the benchmark. This portfolio is built to lag in good times in exchange for protection in bad ones.



7 · Does the optimiser add value? Five weighting methods compared

Same 46 assets, same risk model, same expected returns, five ways to set the weights. This is the test any optimiser has to pass: it must beat naive diversification by enough to justify the estimation risk it takes on. Full code and outputs are in *Portfolio_Optimization.ipynb*.

Method	E[R]	Vol	Sharpe	Max wt	Max risk	Eff. N	Hedges	Verdict
1/N equal weight	9.5%	6.8%	0.98	2.2%	6%	46	35%	Zero estimation error; breaks the $\geq 38\%$ hedge mandate
Long-only max Sharpe	12.4%	5.0%	1.91	22.5%	23%	10	27%	Best on paper; ~ 10 effective bets, below the hedge floor
Minimum variance	3.6%	1.6%	0.53	49.6%	50%	3	75%	Half the book in one hedge; return too low
Risk parity	7.9%	5.0%	1.02	14.0%	2%	24	53%	Competitive without using any valuation views
Final (mandate + RP penalty)	9.4%	6.1%	1.07	5.0%	9%	37	40%	Used: meets every mandate limit

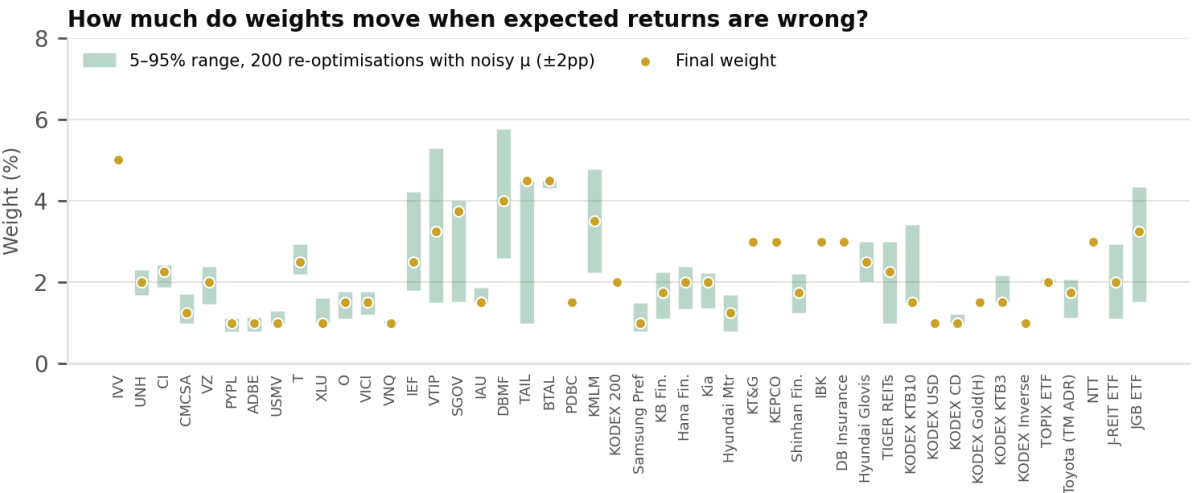


Result. Equal weighting the same 46 assets scores a Sharpe of 0.98, against 1.07 for the final portfolio, and risk parity reaches 1.02 without using a single valuation. So **most of the value came from choosing the 46 assets, not from the optimiser**, consistent with DeMiguel, Garlappi & Uppal (2009). What the optimiser adds is **mandate compliance**: volatility 6.1% vs 6.8% for 1/N, hedges at 40% (1/N only reaches 35%, below the 38% floor), and no position above 5%. The textbook unconstrained Markowitz solution promises a Sharpe of 8.2 but needs 20x gross leverage, mostly shorting one near-riskless asset against another. It is an artefact of estimation, not an opportunity.

Robustness checks

λ (risk-parity penalty)	Sharpe	Vol	Max risk
0	1.12	6.2%	9%
0.25	1.09	6.1%	9%
0.5 (used)	1.07	6.1%	9%
1	1.05	6.1%	9%
5	1.01	6.1%	9%

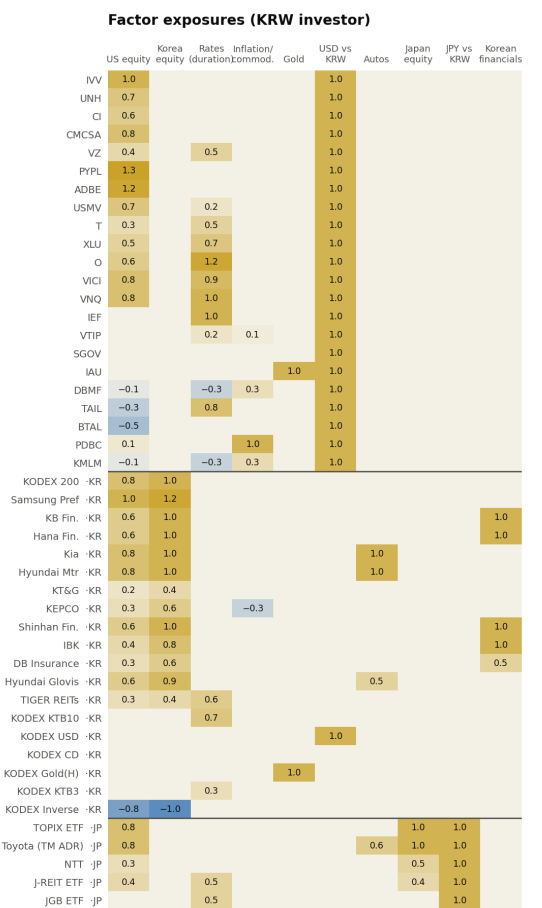
Penalty strength. Moving λ from 0 (pure max-Sharpe inside the mandate) to 5 (near risk parity) changes the Sharpe only from 1.12 to 1.01. The result is not driven by the choice of λ . **Noisy expected returns.** Re-optimising 200 times with ± 2 pp random errors in every μ moves weights by 0.9pp on average (5–95% band). Single-stock weights barely move because their caps bind. The real uncertainty is **the hedge mix**: the bond, TIPS, trend-following and tail-hedge funds swing the most, so those weights should be read as ranges, not point estimates.



8 · Methodology, assumptions and sources

Factor	Vol	Key correlations (assumed)	What loads on it
US equity	16%	−0.35 with USD/KRW; +0.20 with inflation	US stocks, IVV, USMV, REITs; negative for TAIL, BTAL
Korea equity (Korea-specific)	14%	−0.45 with USD/KRW; 0 with US factor by construction	Korean stocks, KODEX 200, K-REITs
Rates (duration return)	7%	−0.30 with inflation; +0.25 with gold	IEF, KTB, TAIL, REITs, Verizon; negative for DBMF
Inflation / commodities	18%	+0.35 with gold	PDBC, DBMF, VTIP; negative for KEPCO
Gold	15%	+0.10 with USD/KRW	IAU
USD vs KRW	9%	Rises when equities fall	Every US-listed asset (unhedged) + KODEX USD futures
Autos	12%	Independent	Kia, Hyundai Motor, Toyota, Hyundai Glovis
Japan equity (Japan-specific)	13%	−0.30 with JPY/KRW (strong yen hurts exporters)	TOPIX ETF, Toyota, NTT, J-REITs
JPY vs KRW	11%	+0.55 with USD/KRW; −0.40 with Korea equity	Every Japanese asset (unhedged), incl. the Toyota ADR
Korean financials	12%	Independent	KB, Hana, Shinhan, IBK; DB Insurance at 0.5

Covariance. $\Sigma = B \cdot F \cdot B^T + D$, where B holds each asset's factor loadings, F is the factor covariance above and D holds stock-specific variance (1–28% vol). US assets load 1.0 on USD/KRW and Japanese assets (incl. the Toyota ADR) 1.0 on JPY/KRW because a Korean investor holds them unhedged. **Expected returns.** Held stocks: CAPM cost of equity + half the valuation gap closed over 3 years. Funds: yield-based or long-run premia (IVV 6.5%, KODEX 200 8.5%, IEF 5.0%, VTIP 4.7%, SGOV 4.2%, gold 4.0%, DBMF/KMLM 5.5%, PDBC 4.5%, TAIL 1.0%, BTAL 2.0%, TOPIX 7.5%, JGB 2.5%, KODEX USD −1.2%, KODEX Inverse −3%). FX rates are assumed to drift 0 (no currency forecast). **Stress tests** apply factor shocks anchored to 2008 and 2022, plus overrides where an asset is known to behave non-linearly (e.g. TAIL, DBMF, BTAL, Samsung in a memory bust, KEPCO in an oil spike).



Rebalancing. Quarterly, or when a weight drifts $\pm 25\%$ from target; valuations re-run each earnings season. A name is sold when it reaches fair value or its thesis breaks (UNH or Cigna margins fall again, Value-up payouts blocked, KEPCO tariffs frozen, VICI rent reset $>15\%$).

Limitations. Volatilities, correlations, factor loadings and stress shocks are the author's capital-market assumptions anchored to history; they were not estimated from a fresh price download. Normal VaR understates fat tails. TAIL's option payoff is non-linear and a linear factor model only approximates it. Samsung (₩150tn) and SK Hynix (₩50tn) net cash, KRW CD yield, J-REIT distribution yield and several normalised ROEs are estimates, and every model shows a sensitivity range for that reason. Weights are rounded to 0.25%.

Sources

US prices and fundamentals: Tickzen stock overviews (MSFT, GOOGL, UNH, JNJ, CMCSA, VZ, PYPL, BMY, CI, C, INTC, LMT, CVS, UPS, BAC, F, ADBE, CRM, QCOM, MRK, FDX, META, PFE), Stock Titan financial summaries (MSFT, GOOGL, UNH, BMY, PYPL, CI, VZ, CRM), ad-hoc-news market reports (Prologis, Realty Income, VICI, Chevron, Altria, Adobe), Prologis and Realty Income 2026 guidance. Korea: Top Star News KRX market reports (Samsung Electronics common and pref, Hyundai Motor, Kia, SK Telecom, Hyundai Mobis, NAVER, SK Hynix, LG Electronics, Hanwha Aerospace), Judal (KB Financial, Hana Financial, Samsung Fire, Samsung C&T, Samsung Life), Chickstock (KT&G, KEPCO), Daum/FnGuide consensus (Samsung, SK Hynix operating profit), Money Today (Samsung Q2 2026 capex), FunETF (TIGER REITs, KODEX USD futures). New in this version: Tickzen (AT&T, Gilead, Medtronic, Humana), AT&T 2026 FCF guidance, Judal (Shinhan, IBK, DB Insurance, Samsung Securities, Hyundai Glovis, POSCO Holdings), Stock Analysis (Toyota, NTT, Honda, Japan Tobacco, Tokio Marine, SMFG, MUFG, Mitsubishi Corp). Macro: Washington Trust weekly review (25 Sep 2026), Trading Economics (KTB and JGB 10-yr), Capital.com (Nikkei, BOJ).